

Hamiltonian Monte Carlo Method for Sampling

Denis Rubtsov

MIPT, Intelligence Systems Department

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Goal of the talk

- ▶ Target: sample from a probability distribution on $\mathbb{R}^d$

$$\pi(x) \propto e^{-f(x)}, \quad x \in \mathbb{R}^d,$$

where f is a potential (or negative log-density).

- ▶ This covers many Gibbs / Boltzmann distributions.
- ▶ Hamiltonian Monte Carlo (HMC) combines
 - ▶ Markov chain Monte Carlo (MCMC),
 - ▶ Hamiltonian dynamics from classical mechanics.
- ▶ Idea: use physics-inspired dynamics to make *long, informed* moves that still preserve the target distribution.

Motivation: why care about sampling?

- ▶ Bayesian inference: posterior sampling for parameters and latent variables.
- ▶ Machine learning:
 - ▶ training and uncertainty estimation in complex models,
 - ▶ approximate integration, model averaging, Bayesian model selection.
- ▶ Physics and chemistry:
 - ▶ molecular dynamics, Ising models, reaction rates, material design.
- ▶ Optimization:
 - ▶ escaping local minima / saddle points,
 - ▶ robust training in non-convex landscapes.

Classical MCMC and its limitations

Random-walk Metropolis / Ball walk

- ▶ Propose

$$\tilde{X}_{i+1} = X_i + \eta V_i,$$

where V_i is isotropic noise (e.g. Gaussian).

- ▶ Accept with probability

$$\alpha = \min \left(1, \frac{\pi(\tilde{X}_{i+1})}{\pi(X_i)} \right).$$

Problems:

- ▶ Step size η must be small to keep acceptance reasonable.
- ▶ In high dimensions, random walk explores space slowly (diffusive behaviour, mixing time often scales at least like d).

Langevin algorithms

- ▶ Use gradient information:

$$X_{i+1} = X_i - \eta \nabla f(X_i) + \sqrt{2\eta} V_i.$$

- ▶ Improvement over pure random walk, but “momentum” is lost after each step.

Hamiltonian dynamics

Consider a particle of unit mass with

- ▶ position $x \in \mathbb{R}^d$,
- ▶ velocity (momentum) $v \in \mathbb{R}^d$.

Define the **Hamiltonian**

$$H(x, v) = f(x) + \frac{1}{2} \|v\|^2,$$

where $f(x)$ plays the role of potential energy.

Hamilton's equations

$$\frac{dx}{dt} = \frac{\partial H}{\partial v} = v, \quad \frac{dv}{dt} = -\frac{\partial H}{\partial x} = -\nabla f(x).$$

The solution at time t starting from (x, v) is denoted

$$\varphi_t(x, v) = (x_t(x, v), v_t(x, v)),$$

the *Hamiltonian flow*.

Key properties of Hamiltonian dynamics

For the dynamics defined by $H(x, v) = f(x) + \frac{1}{2}\|v\|^2$:

- ▶ **Energy conservation:**

$$H(x_t, v_t) = H(x_0, v_0) \quad \forall t.$$

- ▶ **Volume preservation in phase space:**

- ▶ The flow φ_t has Jacobian determinant 1, i.e. it preserves Lebesgue measure on (x, v) space.

- ▶ **Time reversibility:**

$$\varphi_t(x_t, -v_t) = (x_0, -v_0),$$

so dynamics can be run forward or backward.

These properties are crucial: they allow us to build Markov transitions that preserve a desired joint density on (x, v) .

Example

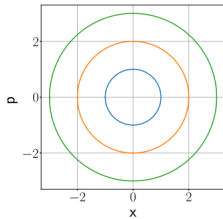
Пример

$$U(x) = \frac{x^2}{2}, \quad K(p) = \frac{p^2}{2}.$$

$$\frac{dx}{dt} = p, \quad \frac{dp}{dt} = -x.$$

$$x(t) = r \sin(t + \varphi),$$

$$p(t) = r \cos(t + \varphi).$$



Extended target distribution

Define a joint density on position x and momentum v :

$$\pi(x, v) \propto \exp(-H(x, v)) = \exp\left(-f(x) - \frac{1}{2}\|v\|^2\right).$$

Then:

- ▶ The conditional distribution of v given x is

$$v \mid x \sim \mathcal{N}(0, I_d).$$

- ▶ Marginalizing over v gives

$$\pi(x) \propto e^{-f(x)},$$

which is our target density.

- ▶ Under exact Hamiltonian dynamics, the density $\pi(x, v)$ is *invariant*:

$$(X, V) \sim \pi \quad \Rightarrow \quad \varphi_T(X, V) \sim \pi \quad \forall T \geq 0.$$

Idealized Hamiltonian Monte Carlo

Algorithm (idealized HMC with exact integration)

1. Start at $X_0 \in \mathbb{R}^d$.
2. For $i = 1, 2, \dots, k$:
 - 2.1 Sample a fresh momentum

$$\xi_i \sim \mathcal{N}(0, I_d).$$

- 2.2 Run Hamiltonian dynamics for time T :

$$(X_i, V_i) = \varphi_T(X_{i-1}, \xi_i).$$

Properties

- ▶ The Markov chain (X_i, V_i) has invariant density $\pi(x, v)$.
- ▶ The marginal chain (X_i) has invariant density $\pi(x)$.
- ▶ No Metropolis correction is needed if we integrate exactly.

Why does idealized HMC preserve $\pi(x, v)$?

Sketch of argument:

- ▶ Start from $(X, V) \sim \pi(x, v) \propto e^{-H(x, v)}$.
- ▶ Along the trajectory (x_t, v_t) ,

$$H(x_t, v_t) = H(x_0, v_0),$$

so the value of e^{-H} is constant on orbits.

- ▶ The flow φ_T preserves phase-space volume (Jacobian determinant = 1).
- ▶ Therefore, the push-forward of π under φ_T is again π .

Conclusion:

- ▶ Any map that is energy-preserving and volume-preserving leaves $\pi(x, v) \propto e^{-H(x, v)}$ invariant.
- ▶ Resampling momentum from $\mathcal{N}(0, I_d)$ between trajectories also preserves the marginal v -distribution.

Convergence for strongly log-concave targets

Assume:

- ▶ f is twice differentiable.
- ▶ f is m -strongly convex and M -smooth:

$$mI \preceq \nabla^2 f(x) \preceq MI \quad \forall x.$$

Then for idealized HMC with appropriate integration time T :

- ▶ The distance between their positions contracts geometrically in Wasserstein distance:

$$W_2(\nu_k, \pi) \leq C(1 - \gamma)^k,$$

for constants C and γ depending on m, M .

- ▶ With $T = \Theta(\sqrt{m/M})$ the number of steps to reach error ε is

$$k = \mathcal{O}\left(\left(\frac{M}{m}\right)^2 \log \frac{1}{\varepsilon}\right),$$

and can be improved to linear in M/m in sharper analyses.

Discretizing Hamiltonian dynamics

In practice we cannot solve Hamilton's ODEs exactly. We use a numerical integrator, typically **leapfrog**:

$$\begin{aligned}v_{t+\frac{\varepsilon}{2}} &= v_t - \frac{\varepsilon}{2} \nabla f(x_t), \\x_{t+\varepsilon} &= x_t + \varepsilon v_{t+\frac{\varepsilon}{2}}, \\v_{t+\varepsilon} &= v_{t+\frac{\varepsilon}{2}} - \frac{\varepsilon}{2} \nabla f(x_{t+\varepsilon}),\end{aligned}$$

where ε is a step size.

Properties of leapfrog:

- ▶ Second-order accurate: error over time $T = L\varepsilon$ is $\mathcal{O}(T\varepsilon^2)$.
- ▶ Time reversible.
- ▶ Symplectic: approximately preserves phase-space volume and a modified Hamiltonian.

Practical HMC algorithm (with Metropolis step)

Input: current position x , step size ε , number of steps L .

1. Sample momentum $p \sim \mathcal{N}(0, I_d)$.
2. Set $(x^{(0)}, p^{(0)}) = (x, p)$.
3. Apply L leapfrog steps with step size ε to obtain $(x^{(L)}, p^{(L)})$.
4. Negate momentum: $\tilde{p} = -p^{(L)}$ (for reversibility).
5. Accept $(x', p') = (x^{(L)}, \tilde{p})$ with probability

$$\alpha = \min \left(1, \exp \left(-H(x', p') + H(x, p) \right) \right).$$

Otherwise keep $(x', p') = (x, p)$.

6. Output new position x' and discard momentum.

Tuning and comparison

Tuning parameters

- ▶ Step size ε :
 - ▶ too large $\Rightarrow$ high discretization error, low acceptance;
 - ▶ too small $\Rightarrow$ expensive trajectories.
- ▶ Number of leapfrog steps L or total integration time $T = L\varepsilon$.
- ▶ Mass matrix (covariance of momentum) can be adapted to geometry.

Advantages over random-walk / Gibbs / Langevin

- ▶ Uses gradients and momentum to move *ballistically* rather than diffusively.
- ▶ Can traverse long distances in state space while maintaining high acceptance.
- ▶ Better scaling in high dimensions for many problems.

Summary

- ▶ HMC augments position x with momentum v and simulates Hamiltonian dynamics to propose distant, high-probability moves.
- ▶ Exact Hamiltonian flow preserves the extended density $\pi(x, v) \propto \exp(-H(x, v))$, hence the target $\pi(x)$.
- ▶ For strongly convex and smooth f , the idealized HMC chain converges rapidly in Wasserstein distance.
- ▶ In practice, leapfrog integration + Metropolis correction yields an efficient and widely used sampler (e.g. in Stan).